

Project Progress Report: Week 1 & Week 2

Project Title: CodeDNA – Biomorphic Code Visualization Engine

Task Domain: Project Planning, Requirements Analysis, & System Design

1. Project Planning and Concept Finalization

During the initial phase of development, the core concept of the project was established. The system is designed to transform Java source code into a biological-style visual organism.

The primary idea is to represent code not as plain text, but as a living structure where each programming element contributes to the shape and behavior of a generated creature.

Key conceptual decisions include:

- Treating the entire program as the body of an organism
- Mapping methods to external extensions (tentacles)
- Representing loops and conditions as dynamic structural elements
- Using complexity to influence mutations and visual variation

This phase ensured that the project direction is clearly defined before implementation begins.

2. Requirements Analysis and System Definition

The second phase focused on identifying the system requirements and defining how the application will operate.

Input Requirements

- A Java source file provided by the user
- File stored in a predefined sample/input directory

Output Requirements

- A generated visual organism
- Output displayed in a browser using HTML and SVG

Functional Requirements

- Read and process Java source files
- Extract key elements such as methods, loops, and conditions
- Analyze structural complexity
- Convert extracted data into visual creature features
- Generate a unique output for each input file

3. System Architecture Design

The system workflow was structured into a step-by-step pipeline to ensure smooth development:

1. File Input Module
 - Reads the Java source file from the system
2. Code Analysis Module
 - Extracts structural elements (methods, loops, conditions, comments)
3. Mapping Module
 - Converts code elements into biological features
 - Example: number of methods → number of tentacles
4. Visualization Module
 - Generates SVG-based graphical representation
5. Output Module
 - Embeds visualization into an HTML file for browser display

4. Workflow Understanding

The overall execution flow of the system is defined as follows:

- User provides a Java file
- System reads and analyzes the code
- Extracted data is mapped to creature structure
- A visual organism is generated
- Output is displayed in the browser

5. Development Summary

By the end of Week 2, the conceptual foundation and system design of the project have been completed. The project is now prepared to move into the implementation phase, starting with environment setup and basic code parsing in the following weeks.